

ALEXANDRE (YIMIN) CHEN

4410 Westheimer Rd. APT 3418A, Houston, TX, 77057

(+1)5415258022 ◊ yimincuo@gmail.com ◊ <https://github.com/AlexandreChern>

EDUCATION

University of Oregon, Eugene

September 2018 - June 2024

Department of Computer Science

Doctor of Philosophy, Computer Science (Focus: GPU Acceleration & Numerical PDEs)

Fudan University, Shanghai

September 2013 - June 2018

Department of Physics

Bachelor of Science, Physics

TECHNICAL SKILLS

Daily Programming Languages

Julia (7 years), Python (8 years), C/C++ (10 years)

Familiar Programming Languages

Fortran (2 years), MATLAB, R, Java, Mathematica

Parallel Processing

CUDA, OpenMP, MPI, Julia Distributed, PETSc

Deep Learning Frameworks & Tools

PyTorch, Tensorflow, Flux.jl, DiffEqFlux.jl, JAX

Software Engineering & Performance

Git, Docker, Flask, GDB, Valgrind, NVIDIA Nsight

Cloud Computing & HPC

Google Cloud, NVIDIA NGC, SLURM

Other Software

Claude Code, LaTeX, Vim, Tmux, VS Code

WORK EXPERIENCE

BP America Inc, Houston TX

June 2024 - Present

HPC Computational Scientist — Center for High Performance Computing

- **GPU Acceleration Engine:** Architected and productionized BP's very first 3D multi-GPU Elastic Finite Difference (FD) elastic Full Waveform Inversion (EFWI) wave propagator in collaboration with NVIDIA team; achieved a **14× single-GPU (H100) speedup** over Intel Sapphire Rapids and over **100× node-level speedup** with near-linear multi-GPU strong scaling up to 8 GPUs.
- **10X Convergence Optimization:** Developed a GPU-accelerated Wavelet Hessian Preconditioned FWI workflow, **slashing iteration counts by 10×** while maintaining comparable or better seismic imaging quality.
- **Asynchronous CUDA Orchestration:** Engineered custom CUDA streams and memory access patterns to perfectly overlap halo-exchange MPI/P2P communications with multi-GPU compute kernels, lifting strict hardware bottlenecks.
- **Multi-Parameter EFWI Development:** Coordinated, delivered, and productionized BP's very first high-performance multi-parameter elastic FWI framework on GPUs, directly optimizing core corporate imaging pipelines to generate value for business.
- **NVIDIA Co-Design Partnership:** Collaborated directly with NVIDIA Devtech and Solution Architecture teams to drive hardware-level code porting optimizations and organize advanced deep-dive CUDA training sessions.

Frontier Development Lab, Mountain View CA

Jun 2022 - Aug 2022

AI for Science Researcher (Sponsored by US Department of Energy - AITO)

- **Physics-Informed Optimization:** Accelerated a physics-based Coulomb Rate-and-State geomechanics model utilizing PyTorch backend optimizers, winning the **"Most Useful Contribution" award** out of 13 global teams.

- **Deep Spatiotemporal Networks:** Conducted hyper-parameter tuning and performance evaluation of deep SCINet models to run automated, massive parallel pipelines for induced seismicity forecasting.

University of Oregon, Eugene, OR

Sep 2018 - June 2024

PhD Graduate Research & Teaching Assistant

- **GPU Kernel Design:** Authored high-performance, matrix-free GPU operators (CUDA) utilizing SBP-SAT methods for multi-dimensional partial differential equations, outperforming NVIDIA's proprietary AmgX library.
- **Geometric Multigrid:** Designed geometric multigrid GPU preconditioners and multi-precision strategies to drastically scale numerical stability for massive 3D linear elasticity problems.
- **Large Scale SEAS Simulations** Designed and developed large scale Simulations of Earthquake Sequences and Aseismic Slip (SEAS) simulation working towards my PhD degree. Reduced years of simulation time from traditional CPU-based direct methods to days from GPU-based iterative methods.

SELECTED PUBLICATIONS & TALKS

Alexandre Chen et. al. Matrix-free SBP-SAT finite difference methods and the multigrid preconditioner on GPUs (*Accepted to ICS 2024*).

Brittany A. Erickson et. al. (2022) Incorporating Full Elastodynamic Effects and Dipping Fault Geometries in Community Code Verification Exercises for Simulations of Earthquake Sequences and Aseismic Slip (SEAS) *Bulletin of the Seismological Society of America*, <https://doi.org/10.1785/0120220066>

Invited Speaker at **SIAM Pacific Northwest Section (2022, 2023):** "Matrix-free Methods for Summation-by-Parts Finite Difference Operators on GPUs."